

of graphics and collection of user inputs, while AR is focussed towards intensive processing of the world, in order to blend with the surroundings naturally and efficiently. But both the systems have to be implemented in real-time with little or no latency.

Both the technologies vary largely with respect to the method of delivery. A standard Virtual Reality system is delivered using a head-mounted display that may look like a boxy set of goggles, that replaces your field of view. This experience is accompanied by sophisticated software and sensors, along with a set of controllers to help you interact with the virtual world intuitively. If we talk about Augmented Reality, it is usually delivered through a pair of smart glasses that project objects to the user. Primitively, it might be simulated on a handheld device, where the augmentation is delivered through the screen. This projection is generated using the feedback received from the camera and other miscellaneous sensors of the handheld device.

Inherently, VR is a technology that is less social, since it encapsulates the user inside the virtual world, isolating him or her from the outside world. On the other hand, Augmented Reality brings both the worlds together hence, it is less isolating and the very nature of AR is to make both the worlds interact. But, the isolation of user may not necessarily mean that VR is less social. With emerging technology, efforts are being made to ensure that VR generates a world that is more social and interoperable with other Virtual Reality systems. VR applications like VRChat, AltspaceVR and Converge are bridging the gap of isolation between the users hence ensuring a more social sphere of virtuality.

Stumbling blocks to the technologies

Even though both the technologies have tons of merits that make these technologies irresistible to tech-enthusiasts, there is no doubt the limitations of these technologies are keeping them from flying off. Some limitations are common to both AR and VR since they have evolved from the common idea of virtualization and share a similar process of evolution while some are specific to the implementation of the technologies. Firstly, both the technologies make use of hardware and software that is quite complex and sophisticated. Hence, it adds an overhead to the project cost, making both the technologies mind numbingly expensive. In addition to this, both the technologies are a victim of complex systems that seem to be underdeveloped for an ideal implementation. Hence, these technologies have the potential to scale heights with proper technical advancements in the future.